SIMATS ENGINEERING
Engineer to Excel

SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Course: Fundamentals of Data Science

Course Code: DSA0404

Slot: A

Faculty: KRISHNARAJ M

Submitted By

Y. Sravan Kumar (192472289)

Lab Programs

1.student_average.py

Code:

student_average.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/FDS lab/student_average.py (3.12.3)

File Edit Format Run Options Window Help

```python
import numpy as np

student_scores = np.loadtxt(
    "Students_data.csv",
    delimiter=",",
    skiprows=1,
    usecols=(1, 2, 3, 4)
)

subjects = ["Math", "Science", "English", "History"]

average_scores = np.mean(student_scores, axis=0)

highest_subject = subjects[np.argmax(average_scores)]

print("Average Scores:", np.round(average_scores, 2))
print("Subject with Highest Average Score:", highest_subject)
```

Output:

IDLE Shell 3.12.3

File Edit Shell Debug Options Window Help

```
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr  9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/FDS lab/student_average.py
Average Scores: [82.  88.  83.2 84.8]
Subject with Highest Average Score: Science
>>>
```

2.Past_month_price_average.py

Code:

```
past_month_price_average.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/FDS lab/past_month_price_average.py (3.12.3)
File  Edit  Format  Run  Options  Window  Help
import pandas as pd

df = pd.read_csv("Saless_data.csv")

average_price = df[df["Month_sales"] == "November 2023"]["Price"].mean()

print("Average Price in November 2023:", round(average_price, 2))
```

Output:

```
IDLE Shell 3.12.3
File  Edit  Shell  Debug  Options  Window  Help
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr  9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/FDS lab/past_month_price_average.py
Average Price in November 2023: 25000.0
>>>
```

3. house_average_price.py

```
house_average.py - C:\Users\Sravan Kumar.Y\AppData\Local\Programs\Python\Python312\FDS lab\house_average.py (3.12.3)
File  Edit  Format  Run  Options  Window  Help
import numpy as np

house_data = np.loadtxt(
    "House_data.csv",
    delimiter=",",
    skiprows=1
)

houses_more_than_4 = house_data[house_data[:, 1] > 4]

average_price = np.mean(houses_more_than_4[:, 3])

print("Average Price of Houses with more than 4 Bedrooms:", round(average_price, 2))
```

Output:

```
IDLE Shell 3.12.3
File  Edit  Shell  Debug  Options  Window  Help
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr  9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\Sravan Kumar.Y\AppData\Local\Programs\Python\Python312\FDS lab\house_average.py
Average Price of Houses with more than 4 Bedrooms: 15500000.0
>>>
```

4.quarterly_sales_numpy.py

Code:

```
quarterly_sales_numpy.py - C:\Users\Sravan Kumar.Y\AppData\Local\Programs\Python\Python312\FDS lab\quarterly_sales_numpy.py (3.12.3)
File  Edit  Format  Run  Options  Window  Help
import numpy as np

sales_data = np.genfromtxt(
    "Quarterly_sales_data.csv",
    delimiter=",",
    skip_header=1,
    dtype=str
)

months = sales_data[:, 1]
sales = sales_data[:, 4].astype(float)

Q1 = 0.0
Q4 = 0.0

for i in range(len(months)):
    if months[i] in ["January", "February", "March"]:
        Q1 += sales[i]
    elif months[i] in ["October", "November", "December"]:
        Q4 += sales[i]

total_sales_year = np.sum(sales)

percentage_increase = ((Q4 - Q1) / Q1) * 100

print("Total Sales for the Year:", round(total_sales_year, 2))
print("Percentage Increase from Q1 to Q4:", round(percentage_increase, 2), "%")
```

Output:

```
IDLE Shell 3.12.3
File  Edit  Shell  Debug  Options  Window  Help
    Python 3.12.3 (tags/v3.12.3:f6650f9, Apr  9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
    Type "help", "copyright", "credits" or "license()" for more information.
>>>
    = RESTART: C:\Users\Sravan Kumar.Y\AppData\Local\Programs\Python\Python312\FDS lab\quarterly_sales_numpy.py
    Total Sales for the Year: 590000.0
    Percentage Increase from Q1 to Q4: 50.0 %
>>>
```

5. fuel_efficiency_analysis.py

Code:

```
fuel_efficiency_analysis.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/FDS lab/fuel_efficiency_analysis.py (3.12.3)
File  Edit  Format  Run  Options  Window  Help
import numpy as np

fuel_data = np.genfromtxt(
    "Fuel_data.csv",
    delimiter=",",
    skip_header=1,
    dtype=str
)

fuel_efficiency = fuel_data[:, 7].astype(float)
make = fuel_data[:, 8]

average_efficiency = np.mean(fuel_efficiency)

mazda_eff = fuel_efficiency[make == "mazda"]
audi_eff = fuel_efficiency[make == "audi"]

mazda_avg = np.mean(mazda_eff)
audi_avg = np.mean(audi_eff)

percentage_improvement = ((mazda_avg - audi_avg) / audi_avg) * 100

print("Average Fuel Efficiency of all cars:", round(average_efficiency, 2), "MPG")
print("Average Fuel Efficiency of Mazda:", round(mazda_avg, 2), "MPG")
print("Average Fuel Efficiency of Audi:", round(audi_avg, 2), "MPG")
print("Percentage Improvement from Mazda to Audi:", round(percentage_improvement, 2), "%")
```

Output:

IDLE Shell 3.12.3

File Edit Shell Debug Options Window Help

```
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr  9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/FDS lab/fuel_efficiency_analysis.py
Average Fuel Efficiency of all cars: 28.67 MPG
Average Fuel Efficiency of Mazda: 31.0 MPG
Average Fuel Efficiency of Audi: 27.0 MPG
Percentage Improvement from Mazda to Audi: 14.81 %
>>>
```

6. customer_purchase.py

Code:

customer_purchase.py - C:\Users\Sravan Kumar.Y\AppData\Local\Programs\Python\Python312\FDS lab\customer_purchase.py (3.12.3)

File Edit Format Run Options Window Help

```python
import pandas as pd

df = pd.read_csv("grocerystore.csv")

total_sales = df["Sales"].sum()

print("Total sales for the grocery store:", round(total_sales, 2))

discount = 0.10
tax = 0.18

price_after_discount = total_sales - (total_sales * discount)

price_after_tax = price_after_discount + (price_after_discount * tax)

print("Final amount after applying discount and tax:", round(price_after_tax, 2))
```

Output:

IDLE Shell 3.12.3

File Edit Shell Debug Options Window Help

```
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr  9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\Sravan Kumar.Y\AppData\Local\Programs\Python\Python312\FDS lab\customer_purchase.py
Total sales for the grocery store: 3850
Final amount after applying discount and tax: 4088.7
>>>
```

7.customer_order_analysis.py

Code:

customer_order_analysis.py - C:\Users\Sravan Kumar.Y\AppData\Local\Programs\Python\Python312\FDS lab\customer_order_analysis.py (3.12.3)

File Edit Format Run Options Window Help

```
import pandas as pd

df = pd.read_csv("order_data.csv")

total_orders = df.groupby("Full Name")["Order Count"].sum()

avg_order_per_product = df.groupby("Items")["Order Total"].mean()

earliest_order = df["Order Date"].min()

latest_order = df["Order Date"].max()

print("Total number of orders made by each customer:")
print(total_orders)

print("\nAverage order total for each product:")
print(avg_order_per_product)

print("\nEarliest order date:", earliest_order)

print("Latest order date:", latest_order)
```

Output:

IDLE Shell 3.12.3

File Edit Shell Debug Options Window Help

```
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr  9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\Sravan Kumar.Y\AppData\Local\Programs\Python\Python312\FDS lab\customer_order_analysis.py
Total number of orders made by each customer:
Full Name
Anjali    3
Priya     3
Rahul     5
Name: Order Count, dtype: int64

Average order total for each product:
Items
Headphones     4000.0
Laptop        75000.0
Mobile        40000.0
Name: Order Total, dtype: float64

Earliest order date: 2023-01-10
Latest order date: 2023-05-18
>>>
```

8. five_most_sold_products.py

Code:

five_most_sold_products.py - C:\Users\Sravan Kumar.Y\AppData\Local\Programs\Python\Python312\FDS lab\five_most_sold_products.py (3.12.3)

File Edit Format Run Options Window Help

```
import pandas as pd

df = pd.read_csv("Electronics_data.csv")

top_5_products = df.groupby("Product_Name")["Price"].sum().nlargest(5)

print("Five Most Sold Products:")
print(top_5_products)
```

Output:

IDLE Shell 3.12.3

File Edit Shell Debug Options Window Help

```
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr  9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\Sravan Kumar.Y\AppData\Local\Programs\Python\Python312\FDS lab\five_most_sold_products.py
Five Most Sold Products:
Product_Name
Laptop          165000
TV               62000
Mobile           45000
Refrigerator     45000
AC               40000
Name: Price, dtype: int64
>>>
```